3) By exchange of solvent:

Colloidal solution of few substances like phosphorous or sulphur is obtained by preparing the solutions in alcohol and pouring them into water. As they are insoluble in water, they form colloidal solution.

$$P \text{ in alcohol} + \text{water} \rightarrow P_{sol}.$$

10.5.3 Purification of colloids

The colloidal solutions due to their different methods of preparation may contain impurities. If they are not removed, they may destabilise and precipitate the colloidal solution. This is called coagulation. Hence the impurities mainly electrolytes should be removed to increase the stabilisation of colloid. Purification of colloidal solution can be done by the following methods.

(i) Dialysis (ii) Electrodialysis (iii) Ultrafilteration.

(i) Dialysis

In 1861, T. Graham separated the electrolyte from a colloid using a semipermeable membrane (dialyser). In this method, the colloidal solution is taken in a bag made up of semipermeable membrane. It is suspended in a trough of flowing water, the electrolytes diffuse out of the membrane and they are carried away by water.

Do you Know? Kidney malfunction results in the building up of electrolyte concentration within the blood to toxic levels.

In the Dialysis, recycling of patient's blood is done through considerable length of seimpermeable tube in an isotonic saline solution.

ii) Electrodialysis

The presence of electric field increases the speed of removal of electrolytes from colloidal solution. The colloidal solution containing an electrolyte as impurity is placed between two dialysing membranes enclosed into two compartments filled with water. When current is passed, the impurities pass into water compartment and get removed periodically. This process is faster than dialysis, as the rate of diffusion of electrolytes is increased by the application of electricity.

iii) Ultrafiltration

The pores of ordinary filter papers permit the passage of colloidal solutions. In ultra filtrations, the membranes are made by using collodion cellophane or visiking. When a colloidal solution is filtered using such a filter,

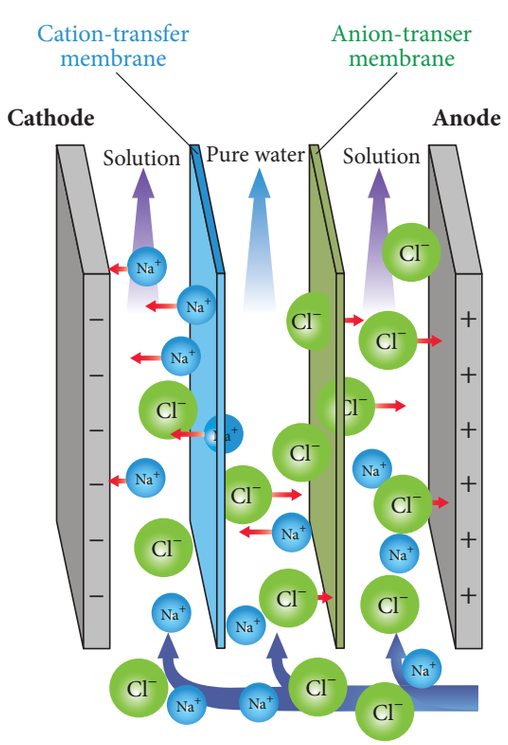


Figure 10.11 Electro Dialysis

XII U10-Surface Chemistry.indd 91 11-01-2021 15:33:43